

## About

These pinouts show what pins are used as configured in the ArPiRobot OS images  
These do not show all possible uses of each pin or all hw interfaces available on the board  
See official pinouts to locate this (official documentation is linked for each board)

## Raspberry Pi Zero W



| Physical  |       |
|-----------|-------|
| 3.3V      | 1 2   |
| I2C1 SDA  | 3 4   |
| I2C1 SCL  | 5 6   |
| GPIO 4    | 7 8   |
| GND       | 9 10  |
| GPIO 17   | 11 12 |
| GPIO 27   | 13 14 |
| GPIO 22   | 15 16 |
| 3.3V      | 17 18 |
| SPI0 MOSI | 19 20 |
| SPI0 MISO | 21 22 |
| SPI0 SCLK | 23 24 |
| GND       | 25 26 |
| I2C0 SDA  | 27 28 |
| GPIO 5    | 29 30 |
| GPIO 6    | 31 32 |
| GPIO 13   | 33 34 |
| GPIO 19   | 35 36 |
| GPIO 26   | 37 38 |
| GND       | 39 40 |

|          |
|----------|
| 5V       |
| 5V       |
| GND      |
| UART0 TX |
| UART0 RX |
| GPIO 18  |
| GND      |
| GPIO 23  |
| GPIO 24  |
| GND      |
| GPIO 25  |
| SPI0 CS0 |
| SPI0 CS1 |
| I2C0 SCL |
| GND      |
| GPIO 12  |
| GND      |
| GPIO 16  |
| GPIO 20  |
| GPIO 21  |

|                               |                              |
|-------------------------------|------------------------------|
| <b>RPi Hat Compatible</b>     | Yes – Full 40 pin            |
| <b>Rpi Camera Compatible</b>  | Yes – requires special cable |
| <b>UART Console</b>           | UART0                        |
| <b>UART Console Baud Rate</b> | 115200                       |
| <b>Reserved Interfaces</b>    | UART0, I2C0                  |
| <b>Number USB</b>             | 1 x USB micro-B OTG          |
| <b>Powered By</b>             | USB micro-B                  |
| <b>Boots From</b>             | micro SD Card                |
| <b>Builtin WiFi</b>           | Yes – 2.4GHz                 |
| <b>Ethernet Port</b>          | No                           |
| <b>CPU Cores</b>              | 1                            |
| <b>RAM</b>                    | 512MB                        |
| <b>GPU Video Decoding</b>     | Yes                          |

<https://www.raspberrypi.com/documentation/computers/ras>

## Raspberry Pi Zero 2 W

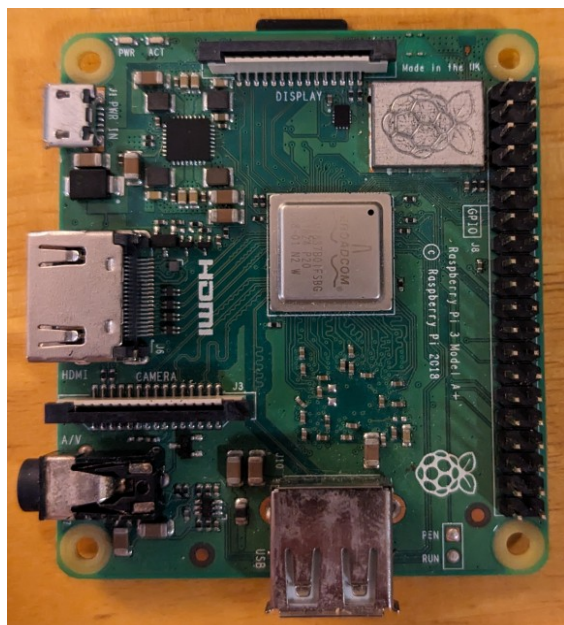


| Physical  |       |
|-----------|-------|
| 3.3V      | 1 2   |
| I2C1 SDA  | 3 4   |
| I2C1 SCL  | 5 6   |
| GPIO 4    | 7 8   |
| GND       | 9 10  |
| GPIO 17   | 11 12 |
| GPIO 27   | 13 14 |
| GPIO 22   | 15 16 |
| 3.3V      | 17 18 |
| SPI0 MOSI | 19 20 |
| SPI0 MISO | 21 22 |
| SPI0 SCLK | 23 24 |
| GND       | 25 26 |
| I2C0 SDA  | 27 28 |
| GPIO 5    | 29 30 |
| GPIO 6    | 31 32 |
| GPIO 13   | 33 34 |
| GPIO 19   | 35 36 |
| GPIO 26   | 37 38 |
| GND       | 39 40 |

|          |
|----------|
| 5V       |
| 5V       |
| GND      |
| UART0 TX |
| UART0 RX |
| GPIO 18  |
| GND      |
| GPIO 23  |
| GPIO 24  |
| GND      |
| GPIO 25  |
| SPI0 CS0 |
| SPI0 CS1 |
| I2C0 SCL |
| GND      |
| GPIO 12  |
| GND      |
| GPIO 16  |
| GPIO 20  |
| GPIO 21  |

|                               |                              |
|-------------------------------|------------------------------|
| <b>RPi Hat Compatible</b>     | Yes – Full 40 pin            |
| <b>Rpi Camera Compatible</b>  | Yes – requires special cable |
| <b>UART Console</b>           | UART0                        |
| <b>UART Console Baud Rate</b> | 115200                       |
| <b>Reserved Interfaces</b>    | UART0, I2C0                  |
| <b>Number USB</b>             | 1 x USB micro-B OTG          |
| <b>Powered By</b>             | USB micro-B                  |
| <b>Boots From</b>             | micro SD Card                |
| <b>Builtin WiFi</b>           | Yes – 2.4GHz                 |
| <b>Ethernet Port</b>          | No                           |
| <b>CPU Cores</b>              | 4                            |
| <b>RAM</b>                    | 512MB                        |
| <b>GPU Video Decoding</b>     | Yes                          |

<https://www.raspberrypi.com/documentation/computers/ras>



## Raspberry Pi 3A+

| Physical  |       |          |
|-----------|-------|----------|
| 3.3V      | 1 2   | 5V       |
| I2C1 SDA  | 3 4   | 5V       |
| I2C1 SCL  | 5 6   | GND      |
| GPIO 4    | 7 8   | UART0 TX |
| GND       | 9 10  | UART0 RX |
| GPIO 17   | 11 12 | GPIO 18  |
| GPIO 27   | 13 14 | GND      |
| GPIO 22   | 15 16 | GPIO 23  |
| 3.3V      | 17 18 | GPIO 24  |
| SPI0 MOSI | 19 20 | GND      |
| SPI0 MISO | 21 22 | GPIO 25  |
| SPI0 SCLK | 23 24 | SPI0 CS0 |
| GND       | 25 26 | SPI0 CS1 |
| I2C0 SDA  | 27 28 | I2C0 SCL |
| GPIO 5    | 29 30 | GND      |
| GPIO 6    | 31 32 | GPIO 12  |
| GPIO 13   | 33 34 | GND      |
| GPIO 19   | 35 36 | GPIO 16  |
| GPIO 26   | 37 38 | GPIO 20  |
| GND       | 39 40 | GPIO 21  |

**RPi Hat Compatible** Yes – Full 40 pin  
**Rpi Camera Compatible** Yes

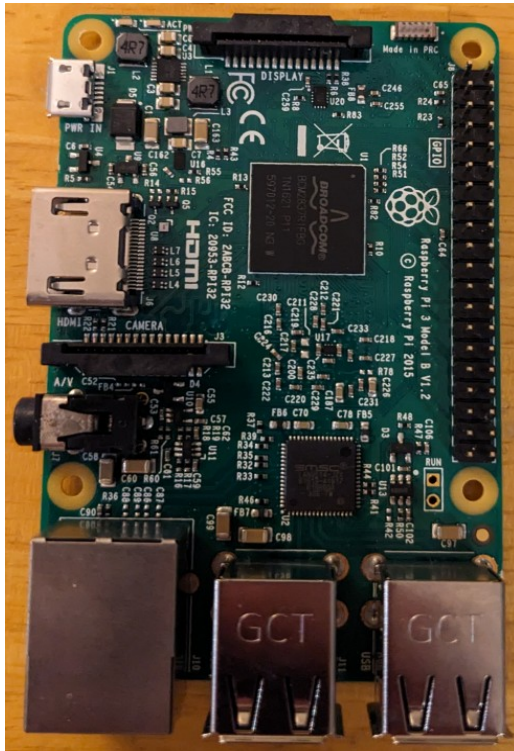
**UART Console** UART0  
**UART Console Baud Rate** 115200  
**Reserved Interfaces** UART0, I2C0

**Number USB** 1 x USB-A  
**Powered By** USB micro-B  
**Boots From** micro SD Card

**Built-in WiFi** Yes – 2.4GHz, 5.0GHz  
**Ethernet Port** No

**CPU Cores** 4  
**RAM** 512MB  
**GPU Video Decoding** Yes

<https://www.raspberrypi.com/documentation/computers/ras>

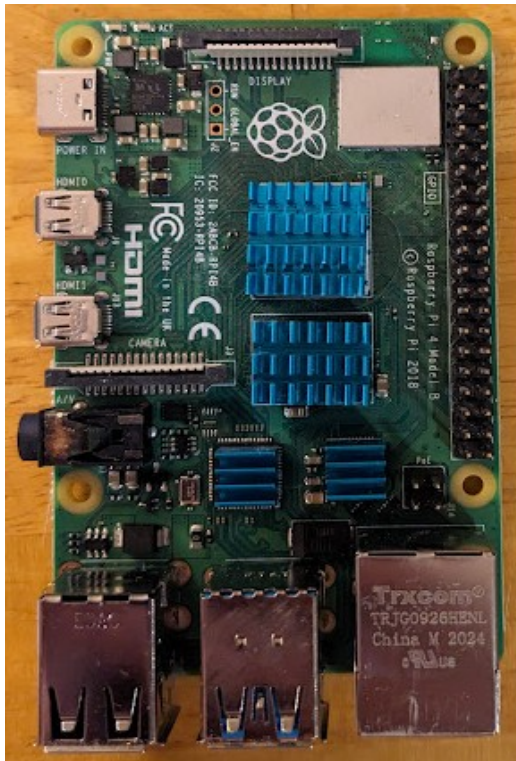


## Raspberry Pi 3B(+)

|           | Physical |          |
|-----------|----------|----------|
| 3.3V      | 1 2      | 5V       |
| I2C1 SDA  | 3 4      | 5V       |
| I2C1 SCL  | 5 6      | GND      |
| GPIO 4    | 7 8      | UART0 TX |
| GND       | 9 10     | UART0 RX |
| GPIO 17   | 11 12    | GPIO 18  |
| GPIO 27   | 13 14    | GND      |
| GPIO 22   | 15 16    | GPIO 23  |
| 3.3V      | 17 18    | GPIO 24  |
| SPI0 MOSI | 19 20    | GND      |
| SPI0 MISO | 21 22    | GPIO 25  |
| SPI0 SCLK | 23 24    | SPI0 CS0 |
| GND       | 25 26    | SPI0 CS1 |
| I2C0 SDA  | 27 28    | I2C0 SCL |
| GPIO 5    | 29 30    | GND      |
| GPIO 6    | 31 32    | GPIO 12  |
| GPIO 13   | 33 34    | GND      |
| GPIO 19   | 35 36    | GPIO 16  |
| GPIO 26   | 37 38    | GPIO 20  |
| GND       | 39 40    | GPIO 21  |

|                               |                             |
|-------------------------------|-----------------------------|
| <b>RPi Hat Compatible</b>     | Yes – Full 40 pin           |
| <b>Rpi Camera Compatible</b>  | Yes                         |
| <b>UART Console</b>           | UART0                       |
| <b>UART Console Baud Rate</b> | 115200                      |
| <b>Reserved Interfaces</b>    | UART0, I2C0                 |
| <b>Number USB</b>             | 4 x USB-A                   |
| <b>Powered By</b>             | USB micro-B                 |
| <b>Boots From</b>             | micro SD Card               |
| <b>Built-in WiFi</b>          | Yes – 2.4GHz, 5.0GHz (5.0G) |
| <b>Ethernet Port</b>          | Yes                         |
| <b>CPU Cores</b>              | 4                           |
| <b>RAM</b>                    | 1GB                         |
| <b>GPU Video Decoding</b>     | Yes                         |

<https://www.raspberrypi.com/documentation/computers/ras>



## Raspberry Pi 4B

|           | Physical |          |
|-----------|----------|----------|
| 3.3V      | 1 2      | 5V       |
| I2C1 SDA  | 3 4      | 5V       |
| I2C1 SCL  | 5 6      | GND      |
| GPIO 4    | 7 8      | UART0 TX |
| GND       | 9 10     | UART0 RX |
| GPIO 17   | 11 12    | GPIO 18  |
| GPIO 27   | 13 14    | GND      |
| GPIO 22   | 15 16    | GPIO 23  |
| 3.3V      | 17 18    | GPIO 24  |
| SPI0 MOSI | 19 20    | GND      |
| SPI0 MISO | 21 22    | GPIO 25  |
| SPI0 SCLK | 23 24    | SPI0 CS0 |
| GND       | 25 26    | SPI0 CS1 |
| I2C0 SDA  | 27 28    | I2C0 SCL |
| GPIO 5    | 29 30    | GND      |
| GPIO 6    | 31 32    | GPIO 12  |
| GPIO 13   | 33 34    | GND      |
| GPIO 19   | 35 36    | GPIO 16  |
| GPIO 26   | 37 38    | GPIO 20  |
| GND       | 39 40    | GPIO 21  |

**RPi Hat Compatible** Yes – Full 40 pin  
**Rpi Camera Compatible** Yes

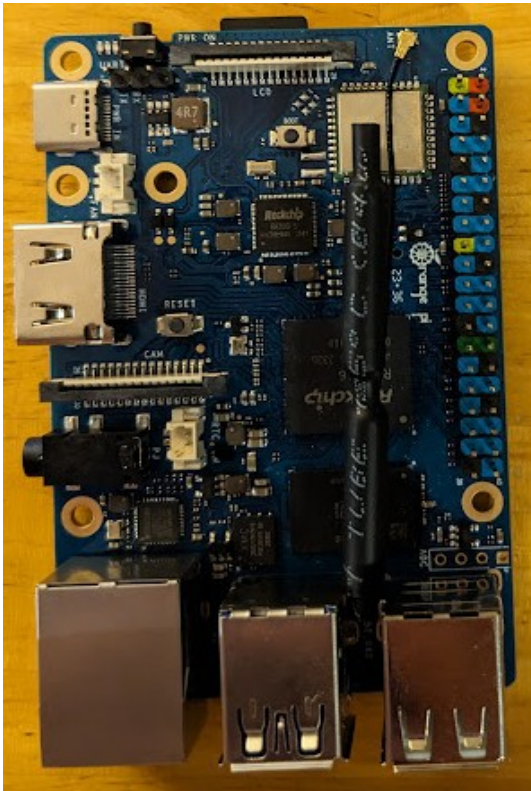
**UART Console** UART0  
**UART Console Baud Rate** 115200  
**Reserved Interfaces** UART0, I2C0

**Number USB** 4 x USB-A  
**Powered By** USB-C  
**Boots From** micro SD Card

**Built-in WiFi** Yes – 2.4GHz, 5.0GHz  
**Ethernet Port** Yes

**CPU Cores** 4  
**RAM** 1GB/2GB/4GB/8GB  
**GPU Video Decoding** Yes

<https://www.raspberrypi.com/documentation/computers/ras>



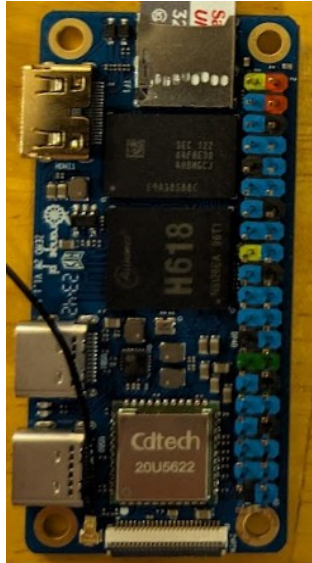
Orange Pi 3B v1.1.1

|           | Physical |          |
|-----------|----------|----------|
| 3.3V      | 1 2      | 5V       |
| I2C2 SDA  | 3 4      | 5V       |
| I2C2 SCL  | 5 6      | GND      |
| GPIO 147  | 7 8      | UART2 TX |
| GND       | 9 10     | UART2 RX |
| GPIO 118  | 11 12    | GPIO 119 |
| GPIO 128  | 13 14    | GND      |
| GPIO 130  | 15 16    | GPIO 131 |
| 3.3V      | 17 18    | GPIO 129 |
| SPI3 MOSI | 19 20    | GND      |
| SPI3 MISO | 21 22    | GPIO 132 |
| SPI3 SCLK | 23 24    | SPI3 CS0 |
| GND       | 25 26    | SPI3 CS1 |
| I2C3 SDA  | 27 28    | I2C3 SCL |
| GPIO 133  | 29 30    | GND      |
| GPIO 124  | 31 32    | GPIO 144 |
| GPIO 127  | 33 34    | GND      |
| GPIO 120  | 35 36    | GPIO 125 |
| GPIO 123  | 37 38    | GPIO 122 |
| GND       | 39 40    | GPIO 121 |

|                               |                              |
|-------------------------------|------------------------------|
| <b>RPi Hat Compatible</b>     | Yes – Full 40 pin            |
| <b>Rpi Camera Compatible</b>  | No – Hardware might be con   |
| <b>UART Console</b>           | UART2                        |
| <b>UART Console Baud Rate</b> | 1500000                      |
| <b>Reserved Interfaces</b>    | UART2                        |
| <b>Number USB</b>             | 4 x USB-A                    |
| <b>Powered By</b>             | USB-C                        |
| <b>Boots From</b>             | micro SD Card                |
| <b>Builtin WiFi</b>           | Yes – 2.4GHz, 5.0GHz         |
| <b>Ethernet Port</b>          | Yes                          |
| <b>CPU Cores</b>              | 4                            |
| <b>RAM</b>                    | 2GB/4GB/8GB                  |
| <b>GPU Video Decoding</b>     | No – Not currently supported |

<http://www.orangepi.org/html/hardWare/computerAndMicr>  
[http://www.orangepi.org/orangepiwiki/index.php/Orange\\_F](http://www.orangepi.org/orangepiwiki/index.php/Orange_F)

*It seems v1.1.1 boards use pinout of v2.1 on 40 pin heade  
 (physical 7 and 22 were different for v1.1?)  
 Thus, this should apply to v1.1.1 and v2.1  
 However, other things on the board were chagned  
 in v2.1 (physical arrangement of some interfaces, baord  
 size, etc)*



|           | Physical |          |
|-----------|----------|----------|
| 3.3V      | 1 2      | 5V       |
| I2C2 SDA  | 3 4      | 5V       |
| I2C2 SCL  | 5 6      | GND      |
| GPIO 269  | 7 8      | UART0 TX |
| GND       | 9 10     | UART0 RX |
| GPIO 226  | 11 12    | GPIO 257 |
| GPIO 227  | 13 14    | GND      |
| GPIO 261  | 15 16    | GPIO 270 |
| 3.3V      | 17 18    | GPIO 228 |
| SPI1 MOSI | 19 20    | GND      |
| SPI1 MISO | 21 22    | GPIO 262 |
| SPI1 SCLK | 23 24    | SPI1 CS0 |
| GND       | 25 26    | SPI1 CS1 |
| I2C3 SDA  | 27 28    | I2C3 SCL |
| GPIO 256  | 29 30    | GND      |
| GPIO 271  | 31 32    | GPIO 267 |
| GPIO 268  | 33 34    | GND      |
| GPIO 258  | 35 36    | GPIO 76  |
| GPIO 272  | 37 38    | GPIO 260 |
| GND       | 39 40    | GPIO 259 |

#### Orange Pi Zero 2W v1.1.1

|                               |                                                                |
|-------------------------------|----------------------------------------------------------------|
| <b>RPi Hat Compatible</b>     | Yes – Full 40 pin                                              |
| <b>Rpi Camera Compatible</b>  | No – connector on edge is not for cameras                      |
| <b>UART Console</b>           | UART0                                                          |
| <b>UART Console Baud Rate</b> | 115200                                                         |
| <b>Reserved Interfaces</b>    | UART2                                                          |
| <b>Number USB</b>             | 1x USB C (USB1 only)                                           |
| <b>Powered By</b>             | USB-C (either)                                                 |
| <b>Boots From</b>             | micro SD Card                                                  |
| <b>Built-in WiFi</b>          | Yes – 2.4GHz, 5.0GHz                                           |
| <b>Ethernet Port</b>          | Yes                                                            |
| <b>CPU Cores</b>              | 4                                                              |
| <b>RAM</b>                    | 1GB/2GB/4GB <b>1.5GB Version NOT SUPPORTED BY ArPiRobot!!!</b> |
| <b>GPU Video Decoding</b>     | No – Not currently supported in software                       |

<http://www.orangepi.org/html/hardWare/computerAndMicrocontrollers/details/Orange-Pi-Zero-2W.html>  
[http://www.orangepi.org/orangepiwiki/index.php/Orange\\_Pi\\_Zero\\_2W#How\\_to\\_install\\_wiringOP](http://www.orangepi.org/orangepiwiki/index.php/Orange_Pi_Zero_2W#How_to_install_wiringOP)

*Note: I2C bus numbers don't match OrangePi pinout diagrams because they don't show up that way in the OS*

*Best I can tell, v1.1 and v1.1.1 are functionally identical*